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28-JAN-2026

DEPARTMENT OF DEFENSE STANDARD PRACTICE

ARMY AIRBORNE RADAR SYSTEM AIRWORTHINESS QUALIFICATION AND VERIFICATION REQUIREMENTS

AMSC N/A FSC 16GP

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FOREWORD

1. This standard is approved for use by the U.S. Army Development Command (DEVCOM) Aviation & Missile Center (AvMC). It is available for use by all Departments and Agencies of the Department of Defense (DoD).

2. This standard provides guidance regarding airworthiness qualification requirements for new or modified radar systems installed on Army aircraft. It establishes the verification methods and requirements to fully satisfy the various airworthiness qualification levels, from safety of flight to full qualification. Airworthiness Systems Readiness Directorate (SRD) of U.S. Combat Capabilities Development Command (DEVCOM) at Redstone Arsenal, Alabama, assesses the airworthiness of the radar systems installed on Army aircraft.

3. Comments, suggestions, or questions on this document should be addressed to Commander, U.S. Army Combat Capabilities Development Command (DEVCOM), Aviation & Missile Center (AvMC), ATTN: FCDD-AMR-MD, 5400 Fowler Road, Huntsville, AL 35898-5000. Since contact information can change, you may want to verify the currency of this address information using ASSIST Online database at https://quicksearch.dla.mil/.

4. Technical questions may be addressed to the following office:

U.S. Army Combat Capabilities Development Command
Aviation and Missile Center
ATTN: FCDD-AMR-AS
Building 4488, Room C-300
Redstone Arsenal, AL 35898-5000
Telephone: Commercial (256) 313-8478
Website: https://www.avmc.army.mil/Directorates/SRD/TechDataMgmt/

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1. SCOPE

1.1 Scope

This military standard provides guidance of airworthiness qualification requirements for new or modified radar systems installed on Army aircraft. This document establishes the verification methods and requirements to fully satisfy the various airworthiness qualification levels, from the safety of flight to full qualification. Airworthiness Systems Readiness Directorate (SRD) of U.S. Combat Capabilities Development Command (DEVCOM) at Redstone Arsenal, Alabama, assesses the airworthiness of the radar systems mounted on the Army aircraft.

Airworthiness qualification consists of analysis, design, operating experiences, and documentation to determine whether the radar's component/subsystem/system is airworthy. The primary purposes of verification and airworthiness qualification are to demonstrate that the radar system can function satisfactorily and safely within the prescribed limits. Airworthiness qualification is required to ensure that the radar is properly integrated into an airworthy Army aircraft.

2. APPLICABLE PUBLICATIONS.

2.1 General.

The publications listed in this section are specified in sections 3, 4, and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

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2.2 Government Publications.

2.2.1 Specifications, Standards, and Handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-1908	Definitions of Human Factors Terms
MIL-HDBK-240-2	Electromagnetic Environmental Effects to Ordnance Guide Part 2, Hazards of Electromagnetic Radiation to Ordnance Testing
MIL-HDBK-704-8	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics 28 VDC (Part 8 of 8 Parts)
MIL-HDBK-781	Handbook for Reliability Test Methods, Plans, and Environments for Engineering, Development, Qualification, and Production
MIL-HDBK-87213	Handbook for Electronically / Optically Generated Airborne Displays

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-E-7016	Electric Load and Power Source Capacity, Aircraft, Analysis of
MIL-L-85762	Lighting, Aircraft, Interior, Night Vision Imaging System (NVIS) Compatible

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1472	Human Engineering Design Criteria Standard
MIL-STD-2525	Joint Military Symbology
MIL-STD-3009	Lighting, Aircraft, Night Vision Imaging System (NVIS) Compatible
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-704	Aircraft Electric Power Characteristics

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MIL-STD-810	Test Method Standard, Environmental Engineering Considerations, and Laboratory Tests
MIL-STD-882	Standard Practice for System Safety

JOINT SERVICE SPECIFICATIONS

JSSG-2010-1	Crew Systems Engineering Handbook
JSSG-2010-5	Crew Systems Aircraft Lighting Handbook
JSSG-2010-7	Crew Systems Crash Protection Handbook
JSSG-2010-11	Crew Systems Emergency Egress Handbook

(Copies of these publications are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government Publications.

The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND (DEVCOM) AVIATION & MISSILE CENTER (AvMC)

ADS-20F-HDBK	Armament and Fire Control System Survey for Army Aircraft
ADS-51-HDBK	Rotorcraft and Aircraft Qualification Handbook
ADS-68-IS	Aircraft Electrical Power Characteristics
ADS-71-SP	Environmental Airworthiness and Qualification Requirements for Electronics, Avionics, and Mission Equipment Installed on Army Aircraft
ADS-74-SP	U.S. Army Aircraft Lighting
AMACC	Army Military Airworthiness Certification Criteria

(Copies of these publications are available online at https://www.avmc.army.mil)

U.S. ARMY MATERIAL COMMAND

AMCP 706-203	Engineering Design Handbook, Helicopter Engineering. Part Three, Qualification Assurance

(Copies of this publication is available online at https://apps.dtic.mil/sti/citations/tr/AD0901657)

DEPARTMENT OF DEFENSE DATA ITEM DESCRIPTIONS (DIDs)

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DI-EMCS-81540	Electromagnetic Environmental Effects (E3) Integration and Analysis Report
DI-EMCS-81541	Electromagnetic Environmental Effects (E3) Verification Procedures (E3VP)

(Copies of these publications are available online at https://quicksearch.dla.mil/)

2.3 Non-Government Publications.

The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the publications are those cited in the solicitation or contract.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS

IEEE C95.1-2019	Standard for Safety Levels with Respect to Human Exposure to Radio Frequency Electromagnetic Fields, 3 kHz to 300 GHz

(Copies of the IEEE document may be obtained from the Institute of Electrical and Electronics Engineers Operations Center, 445 Hoes Lane, Piscataway, NJ 08854-1331 or online at https://www.ieee.orq)

RADIO TECHNICAL COMMISSION FOR AERONAUTICS

RTCA/DO-160	Environmental Conditions and Test Procedures for Airborne Equipment
RTCA/DO-178	Software Considerations in Airborne Systems and Equipment Certification
RTCA/DO-254	Design Assurance Guidance for Airborne Electronics Hardware
RTCA/DO-278	Software Integrity Assurance Considerations for Communication, Navigation, Surveillance and Air Traffic Management (CNS/ATM) Systems

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RTCA/DO-330	Software Tool Qualification Considerations
RTCA/DO-331	Model-Based Development and Verification Supplement to DO-178 and DO-278
RTCA/DO-332	Object-Oriented Technology and Related Techniques Supplement to DO-178 and DO-278

(Copies of all RTCA documents are available online at https://www.rtca.org.)

SOCIETY OF AUTOMOBILE ENGINEERS

SAE/ARP-5580	Recommended Failure Modes and Effects Analysis (FMEA) Practices for Non-Automobile Applications

(Copies of all SAE International documents are available online at https://www.sae.org/.)

2.4 Order of Precedence.

Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Acronyms

AMACC	Army Military Airworthiness Certification Criteria
ADS	Aeronautical Design Standard
AMCP	Army Material Command Pamphlet
AQS	Airworthiness Qualification Standard
AWR	Airworthiness Release
BIT	Built-In Test
CSCI	Computer Software Configuration Item
DVE	Degraded Visual Environment
EED	Electro-Explosive Devices
EID	Electrically Initiated Devices
EMI	Electromagnetic Interference
EMC	Electromagnetic Compatibility

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EMV	Electromagnetic Vulnerability
FMECA	Failure Modes Affect Criticality Analysis
FQT	Formal Qualification Test
HESD	Helicopter Electrostatic Discharge
HFE	Human Factors Engineering
IDD	Interface Design Description
IEEE	Institute of Electrical and Electronic Engineers
IGE	In Ground Effect
JSSG	Joint Service Specification Guide
MOPP	Mission-Oriented Protective Posture
NBC	Nuclear, Biological, and Chemical
NTIA	National Telecommunications and Information Administration
OGE	Out of Ground Effect
PCR	Problem Change Report
PDR	Preliminary Design Review
PESD	Personnel-Borne Electrostatic Discharge
RAM	Reliability, Availability, And Maintainability
RAP	Resonance Assessment Profile (RAP)
RSEC	Radar Spectrum Engineering Criteria
RTCA	Radio Technical Commission for Aeronautics
SAE	Society of Automobile Engineers
SAR	System Assessment Reports
SAWE	Society of Allied Weight Engineers
SDD	Software Design Description
SDP	Software Development Plan
SOF	Safety of Flight
SVD	Software Version Description
STD	Software Test Description
STP	Software Test Plan
STR	Software Test Report
STrR	Software Trouble Report
TEMP	Test & Evaluation Master Plan
TF	Terrain Following

4. GENERAL REQUIREMENTS.

4.1 System Safety.

All component tests and subsystem tests shall be planned and conducted in accordance with the general and detailed requirements of AMACC, Chapter 14. Hazard analyses and safety statements shall be integral and essential factors in the preparation, planning, and conduct of all such tests. Tests shall provide for the assessment of test item hazards associated with further development and testing. System safety reviews shall be an integral part of the program design review.

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4.2 Human Factors Engineering (HFE).

Surveys, ground, and flight tests cited herein shall be used to demonstrate the incorporation of HFE design requirements and criteria in accordance with the general and detailed requirements of AMACC, Chapter 9 "Human Systems Integration," AMACC Appendix A, and MIL-STD-1472 for aircrew meeting the critical body dimensions of 5th percentile female through 95th percentile male, IAW Natick TR 15/007, and the Airworthiness Qualification Standard (AQS). Radar display symbology shall be demonstrated to be in accordance with the general and detailed requirements of MIL-STD-2525. Radar lighting shall be demonstrated to be in accordance with the requirements of MIL-STD-3009 and ADS-74-SP.

4.3 Reliability, Availability, and Maintainability (RAM).

4.3.1 Substantiations and Verifications.

For new or modified components, a combination of substantiation, verification, and demonstration requirements applies to the RAM requirements. Reliability requirements shall be evaluated against the test program results. Maintainability characteristics will be evaluated during Government tests. The reliability, environmental, and test requirements of MIL-HDBK-781 shall be used for guidance.

4.4 Engineering Analyses.

The following engineering analyses shall be conducted.

4.4.1 Electrical Loads Analysis.

An electrical loads analysis shall be conducted in accordance with the requirements of MIL-E-7016. This analysis may be an update to an existing (baseline) electrical loads analysis. Per the requirements of MIL-E-7016, the analysis shall demonstrate the impact of the new system(s) on all aircraft operating modes.

4.4.2 Thermal Analysis.

A thermal analysis shall be conducted to indicate that the thermal requirements of the radar are compatible with the thermal environment of the platform. The thermal analysis shall include worst-case operating conditions, including ground and flight environments, and shall address both hot day and cold weather startups and flight environments. The thermal analysis shall include the thermal loading effects of adjacent equipment installed in the same avionics compartments or areas. Thermal analysis for environmentally cooled Airborne Radar Systems shall address hot-day and cold-weather startup conditions, and hot-day flight operations shall address emergency and or supplemental backup forced air cooling.

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4.4.3 Human Factors Analysis.

The following human factors analyses should be conducted in accordance with the general and detailed requirements of AMACC, Chapter 9, "Human Systems Integration," AMACC Appendix A.

4.4.3.1 Gross Analysis of Tasks.

This analysis shall be performed for all mission scenarios, as defined in the definition section of MIL-HDBK-1908, in which crew task loadings and coordination requirements approach maximum.

4.4.3.2 Displays and Controls Optimization Study.

Trade-off and simulation studies shall be conducted to demonstrate the optimization of control/display relationships in accordance with the general and detailed requirements of MIL-STD-1472. A Controls and Displays Analysis Report should be prepared, using the requirements of MIL-HDBK-87213 as a guide. Target display symbology shall be in accordance with the general and detailed requirements of MIL-STD-2525.

4.4.3.3 Crew Vision.

Analysis and testing shall demonstrate that the vision required for safe flight is maintained during operation with and without night vision goggles.

4.4.3.4 Lighting.

Analysis shall be used to demonstrate that the lighting requirements of MIL-STD-3009 and ADS-74-SP shall be met. The results of the analysis shall address the luminance, chromaticity, and spectral radiance of the display/light sources. The display shall be readable in a combined environment consisting of 10,000 fc diffuse illuminance and the specular reflection of a 2000 fL glare source at rated drive conditions. Display/light sources shall also be compliant with the requirements specified in MIL-STD-3009 and ADS-74-SP for Type I, Class A, Night Vision Goggles. The performance requirements of MIL-L-85762 shall be used as a guide to establish limits on radiant power of emissions within the sensitivity envelope of the Night Vision Imaging System (NVIS). The lighting requirements of JSSG-2010-5 may be used as a guide for this task.

4.4.3.5 HFE Aviator Requirements.

The equipment shall be operated and maintained by 5th percentile female through 95th percentile male, IAW Natick TR 15/007, Army aviator for critical body dimensions with and without protective clothing (Nuclear, Biological, and Chemical (NBC), Mission-Oriented Protective Posture (MOPP) IV, Cold Weather glove, etc.)

4.4.3.6 Emergency Egress.

An emergency egress, IAW MIL-STD-1472, shall be demonstrated by the aircrew members exiting the aircraft within 30 seconds, using only one-half of the exits. The egress requirements of JSSG-2010-11 may be used as a guide for this task.

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4.4.3.7 Crashworthiness.

A crashworthiness test shall be conducted, and the requirements of JSSG-2010-7 may be used as a guide for this task.

4.4.3.8 Workload Analysis.

A workload analysis shall be conducted using the general and detailed requirements of MIL-STD-1472. The analysis shall be done by testing the operator's total cockpit workload during high workload segments of a mission to show that the operator workload is not greater than a "6" using the Bedford Workload Rating Scale (BWRS). The requirements of JSSG-2010-1 may be used as a guide for this task.

4.4.4 Aerodynamic Analysis.

An aerodynamic analysis shall be conducted to determine the aerodynamic effects of any external radar components, including the radome, installed on the helicopter. At a minimum, the radar aerodynamic analysis shall address the impact on aircraft aerodynamic drag (IAW AMACC para. 6.3), handling qualities (IAW AMACC para. 6.25), and flight performance (IAW AMACC para. 6.13).

4.4.5 Structural Analyses.

For the newly designed or modified components, these analyses shall be conducted as described below.

4.4.5.1 Fatigue Analysis.

A fatigue substantiation analysis shall demonstrate the impact of the radar subsystem on component fatigue lives.

4.4.5.2 Loads and Stress Analysis.

This analysis shall be conducted on the radar system, including crash conditions.

4.4.5.3 Vibration, Dynamic Loading, and Frequency Analysis.

The analysis shall show that the newly installed or modified radar subsystem equipment is designed to avoid the airframe critical frequencies. Furthermore, the radar hardware/installation should be designed to minimize vibration-related performance and service life degradation of the radar equipment.

4.4.6 Reliability and Maintainability (RAM) Analysis.

A RAM analysis shall be conducted using test data, calculated estimates, and all qualification testing. The reliability, environmental, and test requirements of MIL-HDBK-781 shall be used for guidance.

4.4.7 Weight and Center-of-Gravity Analysis.

Weight control analysis shall be conducted in accordance with the Society of Allied Weight Engineers (SAWE) Recommended Practice 7 Mass Properties Management and Control for

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Military Aircraft. This analysis shall demonstrate that the aircraft weight and center of gravity remain within the specified acceptable limits with the radar system installed.

4.4.8 Surrounding Equipment and Interoperability.

The analysis shall show that the addition of the radar system will cause no blockage or interference with other systems and demonstrate that any negative effects on performance and integration of surrounding equipment are minimized.

4.4.9 Safety Analysis.

A safety assessment shall be conducted to identify all hazards associated with the system, including the corrective action to either eliminate or reduce all hazards to an acceptable level. Typical hazards encountered with radar equipment include (but are not limited to):

4.4.9.1 Hazards of Electromagnetic Radiation to Personnel (HERP).

A HERP analysis shall be performed in accordance with IEEE STD C95.1-2019 to determine the distance that personnel must remain from the radar while it is operating. This analysis shall also verify that the crew is not vulnerable to the radar while it is operating.

4.4.9.2 Hazardous Materials.

The safety assessment report shall include an analysis of any hazardous materials used in the design and operation of the radar equipment.

4.4.10 Jettison Analysis.

A jettison analysis shall be conducted to ensure that no radar components lie in the jettison path of any external stores.

4.4.11 Installation/Integration Analysis.

This analysis shall describe in detail the methods utilized to determine the optimum locations, design, installation, and test methods for the integration of the equipment on the aircraft.

4.4.12 System Level Lightning Strike Analysis.

Tests or analysis shall be performed IAW AMACC, paragraphs 13.2.7.6 and 13.2.7.9. The sensor control shall survive the effect of a lightning strike as described in the AMACC, paragraphs 13.2.7.5 and 13.2.7.9 document. A lightning strike effects analysis shall demonstrate that a lightning strike (direct or indirect) will not create an unsafe condition and prevent the aircraft from landing.

4.4.13 Software System Safety Hazard Analysis.

The system safety engineers shall perform the hazard analysis process in conjunction with the software development, software test, and Independent Verification and Validation (IV&V) team(s). The software safety engineers shall perform an analysis of requirements, architecture, design, and code and conduct in-depth safety-specific testing IAW MIL-STD-882. A successful software system safety engineering activity is based on a hazard analysis process, a safety-significant software development process, and Level of Rigor (LOR) tasks. An analysis may

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examine in detail the functionality, performance, traceability, and safety implications of a software component, as well as its relationship to other components within the system or equipment.

5. DETAILED REQUIREMENTS.

5.1 Radar System Performance.

Ground tests, flight tests, surveys, and demonstrations shall be used to verify radar integration interfaces, functionality, safe operation, operational capabilities, normal and backup modes, and system performance. Required radar performance parameters shall be as listed in Appendix A of this specification. The standards of this document and TSO-C151B apply to equipment intended to provide the flight crews with both aural and visual alerts to aid in preventing an inadvertent controlled flight into terrain (CFIT) event.

5.1.1 Design Assurance

The Radar hardware and software shall be designed to prevent misleading information and non-announced loss of function.

5.1.1.1 Hardware Compliance

The hardware shall be developed to comply with the test conditions specified in the latest revision of MIL-STD-704 and MIL-HDBK-704-8 at the start of qualification testing for the base hardware. All environmental qualifications should be done in accordance with ADS-71 SP, AMACC, RTCA/DO-160, and MIL-STD-810. The complex electronic hardware shall be designed to comply with RTCA/DO-254 (or equivalent standard).

5.1.1.2 Software Compliance

Software implementing Terrain Following (TF), Degraded Visual Environment (DVE), and other radar functionality shall be developed in accordance with AMACC, Chapter 15. Anomalous behavior of the TF or DVE system software shall be considered a major failure condition, so the software that is developed shall comply with the appropriate software level of the latest RTCA/DO-178 standard. Compliance shall comply with guidance provided in the AMACC Chapter 15 and RTCA/DO-330. RTCA/DO-331, RTCA/DO-332 standards.

5.2 Electromagnetic Environment.

Analyses and tests shall be used to demonstrate that the radar is compatible with the electromagnetic environment as specified below.

5.2.1 Radar Specific Requirements.

Analysis and testing shall demonstrate that the radar design complies with the electromagnetic environmental effects requirements of MIL-STD-461, MIL-STD-464, AMACC Chapter 13, and C95.1-2019 as tailored for the radar system's frequency band.

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5.2.2 EMC Control Plan.

The analysis shall demonstrate how EMI and EMC will be controlled in accordance with the requirements for electromagnetic environmental effects of MIL-STD-461 and MIL-STD-464. The AMACC, paragraph 13.2.8, includes an E3 Control Plan/Integration Analysis per DI-EMCS-81540 to describe the implementation of the E3 interface and performance requirements to evaluate E3 compliance. DI-EMCS-81541 provides Electromagnetic Environmental Effects (E3) Verification Procedures (E3VP) for the data necessary to demonstrate that electromagnetic environment effects (E3) requirements are met.

When required, an EMC Control Plan should show how Electromagnetic Interference (EMI) found during component-level testing should be controlled to ensure the aircraft-level EMC requirements are met. Analysis may be useful in showing that certain measures will reduce the outages in an effort to mitigate all onboard aircraft interactions. Still, it should be backed up by actual testing.

5.2.3 Electromagnetic Interference (EMI).

Testing shall be used to demonstrate that the radar system meets all of the EMI requirements of MIL-STD-461 and AMACC, Chapter 13.1.1. If the radar is flight or safety-critical, RS103 testing shall be conducted to the levels in AMACC, Table 13.1.

5.2.4 Electromagnetic Compatibility (EMC).

Testing shall be used to demonstrate that the radar system, when installed in the aircraft, is electromagnetically compatible with the existing aircraft system equipment in accordance with AMACC, paragraph 13.2.2.

5.2.5 Electromagnetic Vulnerability (EMV).

When installed in the aircraft, the radar system shall survive and meet performance requirements while subjected to an electromagnetic environment composed of field levels per MIL-STD-464 and AMACC paragraph 13.2.4 and field modulations per AMACC, Table 13.1.

5.3 Ground and Flight Testing.

Analyses, tests, and demonstrations shall be conducted using AMCP 706-203 for guidance to substantiate safe and satisfactory radar subsystem operation (in accordance with Appendix A of this document) over the range of flight and environmental conditions.

5.3.1 Ground Tests.

Analysis, tests, and demonstrations shall be conducted using ADS-51-HDBK. These tests shall include all items required to verify flight safety. Required tests include, but are not limited to, the following:

5.3.1.1 Environmental Conditions.

Component-level testing shall be performed according to the Environmental ADS-71 SP in accordance with MIL-STD-810 to verify correct operation throughout the range of conditions specified in the aircraft system specification. Some tests may be tailored as necessary, and

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analysis may be performed to verify compliance with the standards. The equipment should not exhibit damage or degradation of performance during the environmental testing. The test data, analysis, and any simulation results should be available for inspection and review.

5.3.1.2 Electromagnetic Interference (EMI).

Component-level EMI testing shall be conducted in accordance with the requirements for electromagnetic environmental effects of MIL-STD-461. RS103 testing shall be conducted using field levels from MIL-STD-464 and field modulations from AMACC, Table 13.1.

5.3.1.3 Electromagnetic Compatibility (EMC).

Ground EMC testing shall be conducted in accordance with AMACC paragraph 13.2.2. to ensure that the radar system is electromagnetically compatible with other aircraft subsystems prior to flight tests. Testing shall include, but is not limited to, the following:

a. In-flight evaluation of all anomalies experienced during the ground portion of the testing, as well as those equipment and subsystems that cannot be fully tested on the ground.

b. Electrical bonding measurements in accordance with MIL-STD-464, Paragraph 5.11.3. Bonding shall be compliant with MIL-STD-464, paragraph 5.11.

c. Compatibility testing in accordance with a test matrix. Radar systems shall be tested as a source and victim against aircraft navigation and communication systems, flight test instrumentation, electrically initiated devices (EIDs), and all other flight critical systems.

d. Electrically Initiated Device (EID) testing is required due to changes to platform antennas (new, modified, relocated), transmitters, and for new or modified ordnance. This test is required for the first flight with live ordnance on board the aircraft. EIDs defined as having a safety function shall have a minimum of 16.5 dB safety margin. EIDs defined as having a reliability function shall have a minimum of 6 dB safety margin. Compliance shall be verified by testing using the methodologies in MIL-HDBK-240-2

e. Noise floor testing to evaluate the impact of the radar equipment on aircraft communication and navigation equipment.

f. Power quality testing is used to evaluate the impact of the radar equipment on the aircraft power buses.

5.3.1.4 Electromagnetic Vulnerability (EMV).

The radar system, while installed in the aircraft, shall be tested to demonstrate that it can survive and meet performance requirements while subjected to the electromagnetic environment composed of field levels per MIL-STD-464 and AMACC paragraph 13.2.4 and field modulations per AMACC, Table 13.1.

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5.3.1.5 Personnel-Borne Electrostatic Discharge (PESD) Test.

A PESD test shall be performed using MIL-STD-461, test CS118, to determine whether the radar components are safe and reliable when exposed to PESD test environments.

5.3.1.6 Helicopter Electrostatic Discharge (HESD) Test.

A HESD test shall be performed using MIL-STD-464, paragraph 5.8.1, to determine whether the radar components are safe and reliable when exposed to HESD test environments.

5.3.1.7 Electrical Demonstration.

An electrical demonstration shall be conducted to evaluate the radar's compatibility with the aircraft's electrical system.

5.3.1.8 Airframe Vibration Testing.

Vibration testing shall be performed in accordance with MIL-STD-810 with the radar system installed on the aircraft to verify that the radar is compatible with the resonant frequencies of the aircraft, both structurally and operationally.

5.3.1.9 Radar Operations.

Analysis and testing shall verify cockpit procedures utilizing the installed radar control system. The radar-to-aircraft control logic interface shall be demonstrated. Functional checkout of target acquisition/designation subsystem modes (including symbology) shall be conducted.

5.3.1.10 Radar Boresighting.

Analysis and test shall be used to demonstrate boresighting procedures and boresight retention. Particular attention should be paid to the elements of the target acquisition/designation subsystem and boresight accuracy.

5.3.1.11 Aircraft Integration.

Analysis and test shall be used to evaluate the joint functioning of installed subsystems such as fire control computer/weapons processor, target acquisition/designation subsystem, etc. Validate computer software functionality (accuracy, correctness). Use ADS-20-HDBK for guidance on fire control integration.

5.3.1.12 Radar Performance.

When possible, elements of radar performance, as given in Appendix A, shall be tested on the ground before being tested in flight.

5.3.2 Flight Tests.

Flight tests shall be conducted within the design operational flight envelope, as defined by the aircraft operator manual or other appropriate reference. These tests shall include hover-in-ground effect/out-of-ground effect (IGE/OGE), cruise, and VH conditions. The testing shall encompass those items requiring design verification prior to the formal demonstrations. The testing shall include a formal demonstration of radar performance as defined in the system specification. The items examined should include but are not limited to, the following.

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5.3.2.1 Aircraft Flight Performance.

Analysis and test shall be used to determine the effects of the radar subsystem installation on aircraft performance, stability, and control throughout the flight envelope of the aircraft, including hover, low-speed translation flight, take-off and landing, climb, level flight, maneuvering flight, jettisoning, and autorotation.

5.3.2.2 Radar to Aircraft Interface.

Analysis and testing shall be used to verify the following elements of the radar-to-aircraft interface:

a. Boresight retention.

b. Ability to hand over a target from the radar to the aircraft for all target types.

c. Weapons firing. Determine effects of weapons firing on radar system performance, including effects from vibration, smoke blast effects, and debris.

5.3.2.3 Electromagnetic Compatibility (EMC).

Prior to flight tests, flight EMC testing shall be performed in accordance with AMACC, paragraph 13.2.2, to ensure that the radar system is electromagnetically compatible with other aircraft subsystems.

5.3.2.4 Radar Performance.

Analysis and testing shall be used to verify radar performance parameters defined by the system specification. Testing should be conducted with both ground and flight tests, with parameters being tested on the ground first, where possible.

5.4 Other Testing and Verification.

5.4.1 Radar Spectrum Verification.

To verify that the radar system spectrum requirements are met, the radar system shall be tested in accordance with Section 5.5 Radar Spectrum Engineering Criteria (RSEC) of the NTIA Manual of Regulations, 2021 Edition.

5.5 Demonstrations.

5.5.1 General.

In addition to the tests and analysis called out above, demonstration tests can be used to show compliance with system specifications. Demonstrations shall include but are not limited to those listed below.

5.5.2 Electrical Demonstration.

An electrical demonstration shall be conducted to evaluate the effect of the radar system on the aircraft's electrical system. Ripple voltage, transients, harmonics, and power shall be measured

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on the affected busses. As a minimum, the following electrical demonstrations shall be completed:

a. Demonstrate normal aircraft system startup power sequence and verify the Airborne Radar System operates and functions normally.

b. Demonstrate that the Airborne Radar System can satisfactorily operate and function during normal aircraft electrical power bus power switching and failure.

5.5.3 Flight Vibration Demonstration.

If necessary, based on the results of airframe vibration testing or analysis of the radar system's resonant modes, a flight demonstration shall be performed to verify that flight vibration levels do not adversely affect the radar system's performance.

5.5.4 Operating Modes.

At a minimum, an operating mode demonstration shall be completed to demonstrate the satisfactory operation and complete functionality of all radar operating modes, switches, and controls.

5.5.5 Built-in Test (BIT).

Demonstrate that radar BIT capabilities operate and function normally during ground and flight testing. Specifically, verify that the radar startup and status BIT correctly indicate the radar's real operational status.

5.6 Instrumentation and Data Analysis.

Test instrumentation shall be used to record appropriate radar, fire control, weapons processor, and aircraft data to establish qualification test compliance.

5.7 Software Qualification.

5.7.1 Plan for Software Aspects of Certification (PSAC) and Software Accomplishment Summary (SAS).

In-process and final reviews shall be prepared for all new and modified software IAW the Software Development Plan (SDP). The government reserves the right to participate in these reviews.

5.7.2 Software Qualification Testing.

A review of the Software Test Plans (STPs) and Software Test Descriptions (STDs) for conducting all testing to verify compliance with the approved SRS shall be required prior to testing. The Software Test Report (STR) shall be required for review to determine if all airworthiness criteria have been met. All Software Trouble Reports (STrRs) shall be supplied with the STR and any corrective actions that have been taken, including Systems Integration Lab/Hot Bench testing results. A laboratory integration test shall be prepared prior to the flight test, and a statement of flight test suitability shall be issued based on the laboratory results. The

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suitability statement shall identify the pertinent hardware and software configurations/versions in a compatibility matrix as applicable if not part of the Software Version Description (SVD.) A Formal Qualification Test (FQT) shall be prepared with government witnessing to ensure software Safety of Flight has been met.

The complex electronics should be developed, tested, and documented using the latest RTCA/DO-254 (or equivalent standard). The firmware shall be evaluated using the same processes used for software.

5.7.3 Software Safety.

The safety analysis of all software shall be reported in the Safety Assessment Report.

5.7.4 Software Failure Modes Effects and Criticality Analysis (SW FMECA).

For all DAL A and DAL B software, the SW FMECA shall provide an analysis of single-point failures and the resulting impact on the mission, performance safety, and maintainability. The analysis shall be conducted prior to SRR, updated prior to PDR, and delivered to the government for review. The FMECA report shall be prepared using SAE ARP 5580 as a guide. The FMECA is used to determine the following:

a. Root causes of hardware interface anomalies/failures that may impact software

b. Software impacts of hardware interface anomalies/failures

c. Operational impacts of software response to hardware interface anomalies/failures.

5.7.5 Software Documentation.

In addition to the previous documentation listed, the following software documentation should be provided to ensure that software requirements are met, and software verification/validation has been achieved in accordance with DO-278 and AMACC Chapter 15:

a. Software Version Description (SVD)

b. Software Transition Report

c. Software User's Manual

d. Software Design Description

e. Interface Design Description

f. Interface Control Document

g. Interface Requirements Specification

h. Software Requirements Specification

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i. System/segment Design Document

j. System/Segment Specification

k. Source Code

5.7.6 Software Test.

All new or modified software should be extensively tested in the Software Integration Lab/Hot Bench to verify that armament system functionality and safety have not been inadvertently degraded. This test will include, but not be limited to, verification of weapons management functions, no inadvertent launch/firing, and veracity of weapon inhibits, limits, and interrupts (WILIs) on the Weapons Software build.

5.8 Electrical Environment.

Analyses and tests shall demonstrate that the radar is compatible with the power-quality environment.

5.8.1 Radar Specific Requirements.

Analysis and testing shall demonstrate that the radar design complies with the aircraft's electrical and environmental requirements of MIL-STD-704, MIL-HDBK-704-8,and ADS-68-IS.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Subject term (keyword) listing.

Signature Aperture
Silent Knight
Multimode
Terrain
Terrain Avoidance
Terrain Following
Skin Paint
Maritime
Ground Mapping
Weather
Windshear
Turbulence

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APPENDIX A

Radar Parameters.

A.1 Scope.

This Appendix is intended as a general guide for persons developing and reviewing detailed radar system specifications. Listed here are generic (although biased towards airborne radar) requirements that should be defined in the detailed item specification and tested as part of the System Qualification process. Not all parameters will be applicable to all radar designs, and this list should be tailored to the particular operational requirements of the radar system. Conversely, it would only be possible to list some possible detailed requirements, so this list is meant only as a guide rather than a comprehensive reference.

Due to the ever-evolving and wide variety of radar technologies, it would be challenging to define performance parameters for each type of design and each application of that design. Instead, this Appendix defines general radar performance parameters independent of system implementation. To accomplish this, several generalizations are made:

a. No distinction is made between scanning and tracking modes since the modes designed for the radar depend on the radar's operational requirements.

b. No distinction is made between measured parameters and calculated parameters. Calculated parameters will still have accuracies associated with measurement noise, etc., and so are lumped together with measured parameters for simplicity. Also, the set of calculated and measured parameters will vary between radar designs, depending on cost, size, and complexity of design.

c. Mechanical details such as slew rates are not included. Rather, event timelines are defined. Parameters such as slew rate would likely be included in the detailed specification, but only to the extent required to meet timeline requirements.

Note that each of these parameters may be dependent on the radar's operating mode. In this case, the system specifications and test plans should contain matrices of those parameters versus operating mode.

A.2 Radar Performance Parameters.

A.2.1. Maximum radial range, maximum azimuth sweep, and maximum elevation sweep.

A.2.2. Radial range, azimuth, and elevation accuracy.

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A.2.3. Radial, cross-range, and elevation velocity accuracy. Although cross-range and elevation velocities are calculated, they are included here since they depend on measured values, as well as the quality of the scan-to-scan correlation, noise measurement, etc. Having a single accuracy specification makes this item less implementation-specific.

A.2.4. Minimum detectable target size.

A.2.5. Probability of detection (Pd). It is measured over the set of all detectable targets.

A.2.6. Probability of reacquisition. This is the Probability of reacquiring a previously detected target. This is a special case of item 5, applicable only for radars that optimize for reacquisition (for example, when tracking high-priority targets).

A.2.7. Probability of False Detection (Pfd). This is the Probability of falsely detecting a target when none is present.

A.2.8. Probability of correct classification (Pcc). This is only applicable if the radar performs target classification. The specification should contain tabulated values for each type of target the radar is supposed to classify. For consistency and ease of tabulation, Pcc is defined as the Probability of correctly classifying a detected target. It is implicitly conditional upon Pd since the Probability of correctly classifying a non-detected target is zero. The detailed specification may also define different levels of classification, ranging from simple determination of vehicle type to exact identification of the vehicle.

A.2.9. Maximum number of targets that can be tracked at a time, if applicable to the radar design.

A.2.10. Priority Schemes. This defines the method (or methods) of prioritizing targets when multiple targets are being tracked.

A.2.11. Target (or Differential) Resolution. This is defined as the radar's ability to distinguish multiple closely spaced targets. Two different definitions are given below. The applicable definition should be used.

A.2.11.1 Ideally, the target resolution would be specified in each orthogonal dimension, as given below:

A.2.11.1.1. Differential range resolution: The minimum radial distance of two objects must be separated to be detected as distinct targets by the radar, assuming both targets have the same azimuth and elevation.

A.2.11.1.2. Differential azimuth resolution: The minimum cross-range distance between two objects must be separated to be detected as distinct targets by the radar, assuming both targets have the same radial range and elevation.

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A.2.11.1.3. Differential elevation resolution: The minimum vertical distance of two objects must be separated to be detected as distinct targets by the radar, assuming both targets have the same radial range and azimuth.

A.2.12. In the event that the radar design does not allow for such orthogonal resolution measurements, the resolution requirements should be specified as:

A.2.12.1. Differential range resolution (non-orthogonal): The minimum radial distance of two objects must be separated to be detected as distinct targets by the radar, given a specified minimum separation value in azimuth and elevation.

A.2.12.2. Differential azimuth resolution (non-orthogonal): The minimum cross-range distance of two objects must be separated to be detected as distinct targets by the radar, given a specified minimum separation in radial range and elevation.

A.2.12.3. Differential elevation resolution (non-orthogonal): The minimum vertical distance of two objects must be separated to be detected as distinct targets by the radar, given a specified minimum separation in radial range and azimuth.

A.2.13. Timelines. Due to the uncertainty in time delays caused by aircraft software processing, data transmission, etc., the timelines defined below measure only the time it takes to get the data directly from the radar. This allows for a clean analysis of bandwidth and latency requirements. The radar system designer should include a method of directly measuring these delays to avoid inaccuracies caused by attempting to measure them indirectly over the aircraft system bus.

A.2.13.1. Detection: Time required to detect a new (not seen before) target.

A.2.13.2. Reacquisition: Time required to reacquire a previously detected/tracked target that has been dropped. This is applicable to radars that optimize reacquisition. For other radars, this may be the same as the detection time.

A.2.13.3. Automatic Handover: The time required from the time the target is detected until it is handed over to an external bus. This assumes that targets are handed over automatically, as they are detected, without operator intervention.

A.2.13.4. Manual Handover: Time required to hand over an already detected/tracked target. This applies when the radar does not automatically hand over the target but waits for an operator to pick a target to hand over. This is the time between the radar receiving the operator input and the handover of the target to an external bus. This does not count delays introduced by an operator input device, e.g., a touch panel.

A.3 Maximum Operating Envelope and Operation in Degraded Conditions.

The following items should be specified in the detailed item specification:

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A.3.1. Flight Conditions

A.3.1.1. Maximum yaw, pitch, and roll angle for which the radar system will continue to meet its performance requirements.

A.3.1.2. Maximum yaw, pitch, and roll velocities for which the radar system will continue to meet its performance requirements.

A.3.1.3. Maximum yaw, pitch, and roll accelerations for which the radar system will continue to meet its performance requirements.

A.3.2. Aircraft Alignment.

A.3.2.1. Physical. This defines the required accuracy of the physical alignment of the radar to the airframe in order to meet radar performance requirements.

A.3.2.2 Inertial. This defines the required accuracy of the inertial reference between aircraft and radar.

A.3.3. Clutter environment. Tabulated clutter conditions versus percent degradation in radar performance.

A.3.4. Weather. Weather conditions under which the radar will continue to meet its performance requirements.

A.3.5. Obscurants. A tabulated value showing the levels of different obscurants versus the percent degradation in performance should be provided.

A.3.6. Threat environment. The specification should define the types of threats the radar might encounter and describe the radar's performance in each condition. This includes passive threats such as decoys or obscurants and active threats such as jammers. The specification should take into account current and forecasted threats at the time it is written.

A.4 Radar Subtypes/Modes.

This Appendix, up to this point, applies to radars that detect and track air and ground targets. However, the radar may have special modes or be of an entirely different design, for which the above guidance is not relevant. An attempt is made below to give guidance about what should be written into the specification for these special modes or radars.

A.4.1. Terrain Following. Radars for the following terrain, as an aid to navigation, are moderately specialized but have some common characteristics that should be included in the detailed specification.

A.4.1.1. Maximum range, azimuth, and elevation.

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A.4.1.2. Minimum safe flying height and speed, in tabular form if appropriate.

A.4.1.3. Types of objects identified (foliage, buildings, wires, etc.), and, very importantly, whether the radar identifies these objects itself or relies on an external database of previously surveyed data as well as in real-time target detection/classification applications. The parameters in paragraph A.2 should be general enough to cover the case of imaging radars performing target detection and tracking.

A.4.2. Weather Radars. When used as navigational aids, the following parameters should be required as a minimum in the detailed specification.

A.4.2.1. Maximum range, azimuth, and elevation.

A.4.2.2. Ability to identify precipitation, both type and severity, along with the associated accuracies.

A.4.2.3. Ability to identify severe climatic activity, such as high winds or other dangerous conditions, along with the associated accuracies.

A.4.3. Weather radars shall be tested during flight to verify the detection of hazardous weather for the specified range.

A.4.4. Radar ground mapping shall be tested during flight to verify the ability to display terrain type and to locate targets of a specified size at the specified range.

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CONCLUDING MATERIAL

Custodians:
Army – AV

Preparing activity:
Army - AV
(Project 16GP-2023-003)

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